

MINTS: Nucleotide correspondence and control feasibility in genomic transformer interventions

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Genomic transformers split DNA into variable-length tokens, so equal activation shapes can conceal interventions at different nucleotide positions. MINTS implements explicit nucleotide-span and vocabulary correspondence, checks controlled-edit feasibility, and separates trained-readout effects from native model predictions and biological binding evidence. Six of ten historical TATA patching pairs fail correspondence. A repaired discovery-selected CTCF head changes a fixed trained readout by 0.0157 decision-score units on 64 held-out sequences in 45 clusters, with a 95% block interval of [0.0066, 0.0251]. The TATA estimate remains inconclusive, and the selected CTCF head’s measured mean base density is near background. A separately frozen mapped-query study retains 61 natural-variant scenarios in 60 genomic clusters. Variant and sham prediction divergences average 0.013053 and 0.029018 nats; their paired contrast is -0.015966, with a 95% cluster interval of [-0.024454, -0.009146]. Larger sham shifts fail the positive-sensitivity gate and stop native head search. A post-study correspondence checker reproduces all 438 saved queries without new model scoring or changed eligibility. Two limited proofs explain incompatible control constraints. Public code and sequence-free archived evidence reproduce numerical reporting. The contribution is a bounded computational validity audit. Residual segmentation effects, untested cross-model sensitivity and incomplete biological measurement provenance prevent biological binding-causality claims.

The correspondence certificate checks declared span maps and constraints, allowing shifted indices and unequal token counts. It does not establish tokenizer provenance or isolate segmentation elsewhere. Original memberships, endpoints, timestamps and stopped studies remain intact. Numerical reproduction uses saved clusters; fresh inference requires upstream resources under their own terms.

Code: <https://github.com/ArjunCodess/MINTS/tree/recomb-submission-package>. Reproduction data: <https://doi.org/10.5281/zenodo.23267982>.